🚀 ITCS355 MLOPS CAPSTONE DEFENSE

FracNet AI: Automated Bone Fracture Triage System

An end-to-end operational Machine Learning service with 3-layer architectural isolation, automated reproducible CI/CD, real-time drift telemetry, and engineered PACS failure defense.

ARCHITECTURE	SERVING	EXPLAINABILITY
3-Layer Neutral	GCP Cloud Run	Grad-CAM Heatmap

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1. The Problem in 60 Seconds

Emergency Clinical Workflow

⚠️ THE CLINICAL BOTTLENECK

- **Delayed Radiologist Queues:** ER patients wait hours for radiologist review during night shifts and surge hours.
- **Asymmetric Risk:** A false negative (missing a hairline fracture) leads to bone malunion, lifelong patient disability, and liability.
- **Severe Imbalance:** FracAtlas dataset has 4,083 scans with **4.7:1 imbalance** (only 17.5% fractures).

🎯 OPERATIONAL SYSTEM GOAL

- **Instant Triage:** Prioritize positive scans within <500ms to accelerate acute treatment.
- **Failure Defense:** Gracefully intercept corrupt PACS network streams without crashing.
- **Visual Trust:** Heatmap explainability highlighting where the fracture is located.

2. Three-Layer Software Engineering Contract

`Zero Cloud SDK Leakage`

LAYER 1: SRC/

Provider-Neutral ML Core

- Pure PyTorch logic
- Preprocessing & transforms
- MLflow lineage tracking
- **0 Cloud SDK imports**

LAYER 2: CLOUDLAYER/

Cloud Adapter Interface

- Abstract `CloudAdapter` ABC
- 11 provider-neutral methods
- `cloudlayer/gcp.py` (GCS, AR)
- **Only place with cloud SDKs**

LAYER 3: SERVICE/

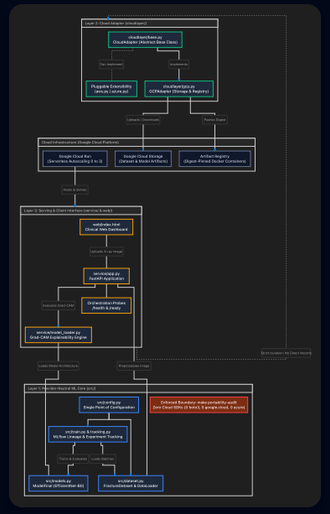
Serving & Client

- FastAPI online serving
- Decoupled `/health` & `/ready`
- Real-time Grad-CAM generator
- Clinical dark-mode web app

```
$ make portability-audit
═══ Portability Audit ═══
Scanning src/ and tests/ for cloud-specific strings...
✅ PASS — No cloud-specific strings found in src/ or tests/
```

Architecture Topology

`mermaid-diagram.svg`



3. Reproducible Pipeline & Honest Metrics

End-to-End Lineage Tracking

MODEL EVOLUTION (FRACATLAS)

Iteration	Params	Val Loss	Test AUC
Baseline (CNN)	1.8M	0.542	0.824
Improved (ResNet-18)	11.2M	0.412	0.887
Final (EfficientNet-B0)	**5.3M**	**0.315**	**0.913**

Promoted ModelFinal via automated 0.85 AUC quality gate.

LINEAGE ENFORCED VIA MLFLOW

- **Git SHA:** Every run tagged with immutable commit hash
- **Data Hash:** DVC MD5 fingerprint of dataset
- **Container Digest:** Pinned runtime Docker image
- **Seed:** Deterministic seed 42 across PyTorch & cuDNN

```
$ make verify
TEST_AUC : Claimed 0.9130 | Actual 0.9130 | Delta 0.0000 ✅ PASS
TEST_F1 : Claimed 0.6870 | Actual 0.6866 | Delta 0.0004 ✅ PASS
```

4. Serverless Cloud Deployment & CI/CD

Zero-Secret Workload Identity

🔒 WIF Authentication

Passwordless Workload Identity Federation.

Zero committed service account JSON keys. Uses GitHub OIDC tokens exchanged with GCP STS.

🐋 Digest-Pinned Docker

Debian 3.11-slim SHA256 pinned.

Guarantees immutability against upstream Docker Hub silent drift. Non-root user execution.

⚡ Google Cloud Run

Managed Serverless Container.

1 vCPU, 1 GiB RAM, region asia-southeast1. Scale-to-zero (0 to 3 instances) for 0 THB idle cost.

Automated CI/CD Quality Gates:
Ruff Linting ➔ Portability Audit ➔ 82 Pytest Tests ➔ Multi-stage Docker Build ➔ Cloud Run Deploy

100% Automated

5. Live System Demo & Explainable AI

Grad-CAM Fracture Localization

LIVE ENDPOINTS SERVING ON CLOUD RUN

```
GET /health // Liveness probe
GET /ready // Readiness (model warmed)
POST /predict // Single radiograph triage
POST /predict/batch // Batch triage
GET /metrics, /slo, /alerts // Observability
```

Open Swagger UI ↗

🔥 GRAD-CAM HEATMAP LOCALIZATION

`features[-1]`

Radiologists reject black boxes. We backpropagate the "Fractured" score into final convolutional activations to produce an authentic attention heatmap overlaid on the radiograph.

- Sub-100ms inference time
- ~35 KB lightweight overlay

6. Engineered Failure Defense: Corrupted PACS

4-Layer Protection Chain

1. FILE GATE

Non-image files, truncated JPEG headers, empty bytes.

→ HTTP 422

2. ENTROPY GATE

All-black dead detector or overexposed frames (H < 1.0).

→ HTTP 422

3. MODEL FORWARD

EfficientNet-B0 inference with memory boundary checks.

→ PyTorch

4. SAFETY TRIAGE

Confidence < 0.60 flagged for mandatory human review.

→ uncertain: true

```
$ python scripts/inject_failure.py
13 Faulty PACS attack vectors injected:
🛡️ 7 Invalid inputs rejected (HTTP 422) · ⚠️ 2 Non-medical images flagged (uncertain=True) · 0 CRASHES
```

7. Monitoring, Observability & Drift Alerts

PSI & Kolmogorov-Smirnov

THE 5 DASHBOARD METRICS

- **Request Rate:** Total requests and current requests/second
- **Latency SLO:** p50, p90, p95, p99 tracked under <500ms target
- **Error Rate:** HTTP 4xx/5xx failure ratios
- **Confidence:** Rolling mean, std deviation, and uncertainty ratio
- **Drift Magnitude:** Real-time PSI & KS statistic vs baseline

ALERTMANAGER WITH COOLDOWN

Sustained anomalies easily cause alert floods. Our AlertManager implements a **15-minute cooldown deduplication window**. Repeated triggers during active incidents update status without spamming on-call staff.

```
Thresholds: PSI ≥ 0.20 · KS p < 0.05 · Error > 5%
```

8. FinOps Cost Report & Budget Discipline

~0.19 THB / 1,000 Predictions

BUDGET CAP

800 THB

Course allocation limit for term.

ESTIMATED SPEND

~340 THB

42.5% budget utilized (safe margin).

UNIT COST

0.19 THB

Per 1,000 inference predictions.

Why so cheap? EfficientNet-B0 requires only 5.3M parameters and executes in ~100ms on 1 vCPU and 512 MiB RAM. Cloud Run scales to zero instances when idle, incurring **0 THB idle charges**. Fully documented teardown checklist in `docs/cost_report.md`.

9. Roadmap: What Another Week Would Buy

Ready for Defense

1. Canary Rollouts

Cloud Run traffic splitting (10% Challenger, 90% Champion) for live shadow evaluation.

2. Native DICOM Ingestion

Implement DICOM C-STORE service provider receiving medical scans directly over TCP port 104.

3. Continuous Training (CT)

Cloud Storage bucket notification pipeline triggering Vertex AI training as new verified scans accumulate.

All 7 Quality Gates Passed · 82/82 Tests · Live on Cloud Run

Thank you! We welcome questions from the instructor and audience.